

Nuclear Fuel Performance

NE 533: Spring 2026

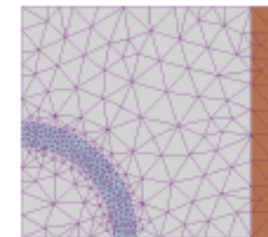
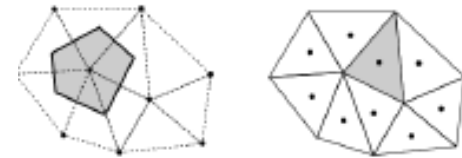
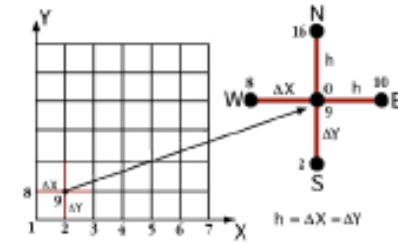
Last Time

- Axial temperature profile
- Thermal operating limits
 - CHF and DNBR
- Phonon-based thermal conductivity
- Numerical discretization
- Forward/backward Euler

SPATIAL DISCRETIZATION

Spatial resolution

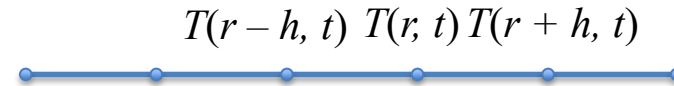
- To numerically solve in space, we need to discretize
 - Finite difference
 - convert differential equations into a system of equations that can be solved by matrix algebra techniques
 - Finite volume
 - volume integrals in a partial differential equation that contain a divergence term are converted to surface integrals, using the divergence theorem. These terms are then evaluated as fluxes at the surfaces of each finite volume
 - Finite element
 - subdivides a large system into smaller, simpler parts that are called finite elements, the equations that model these finite elements are then assembled into a larger system of equations that models the entire problem



Finite Difference

- The finite difference method solves on a grid and uses numerical derivatives
- Derivatives are approximated by differences
- Boundary conditions must have either a fixed T or dT/dr
- Typically restricted to handle rectangular shapes
- Once you compute the time derivative, you can use either forward or backward Euler to march through time

$$\rho c_p \frac{\partial T}{\partial t} = \frac{1}{r} \frac{\partial}{\partial r} \left(r k \frac{\partial T}{\partial r} \right) + Q$$



$$\dot{T}(r, T) = \frac{1}{\rho c_p} \frac{1}{r} \frac{\partial}{\partial r} \left(r k(T) \frac{\partial T(r, T)}{\partial r} \right) + \frac{1}{\rho c_p} Q(r)$$

$$q = r k(T) \frac{\partial T(r, t)}{\partial r} = \frac{r k(T(r, t))}{2h} (T(r+h, t) - T(r-h, t))$$

$$\dot{T}(r, t) = \frac{1}{\rho c_p} \frac{1}{r} \frac{\partial q}{\partial r} + \frac{1}{\rho c_p} Q(r) = \frac{1}{\rho c_p} \frac{1}{2h r} (q(r+h, t) - q(r-h, t)) + \frac{1}{\rho c_p} Q(r)$$

Finite Volume

- Discretize the domain by subdomains
 - Domain size h
 - We place points in the subdomain centers and on either boundary
- The finite volume method balances fluxes across the boundaries of your divided subdomains
- Integrate our PDE across the subdomain
- Evaluate the integral using a linear approximation of the variable
- Restricted to flux boundary conditions, often used in flow-type problems

$$\rho c_p \frac{\partial T}{\partial t} = \frac{1}{r} \frac{\partial}{\partial r} \left(r k \frac{\partial T}{\partial r} \right) + Q$$



$$\frac{d}{dx} k \frac{dT}{dx} + q = 0$$

$$\int_a^{a+h} \frac{d}{dx} k \frac{dT}{dx} + q \, dx = 0$$

$$k \frac{dT}{dx} \Big|_{a+h} - k \frac{dT}{dx} \Big|_a + qh = 0$$

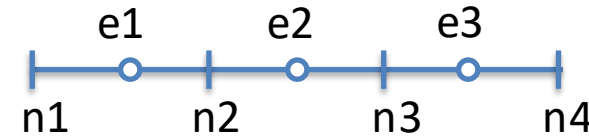
$$\frac{T_{i+1} - T_i}{h_2} - \frac{T_i - T_{i-1}}{h_1} + q \frac{h_2}{k} = 0$$

$$T_i = \frac{h_1 h_2}{h_1 + h_2} \left(\frac{T_{i+1}}{h_2} + \frac{T_{i-1}}{h_1} + q \frac{h_2}{k} \right)$$

Finite Element

- In the finite element method, we interpolate the variable using nodal values and integrate over elements
- Systematically recombine all sets of element equations into a global system of equations for the final calculation
- Write the strong form of the equation, rearrange to get zero on the right-hand side, multiply by the test function, integrate over the domain, yielding weak form
- The **strong form** states conditions that must be met at every material point, whereas **weak form** states conditions that must be met only in an average sense
- Finite element works for any geometry and any boundary condition

$$\rho c_p \frac{\partial T}{\partial t} = \frac{1}{r} \frac{\partial}{\partial r} \left(r k \frac{\partial T}{\partial r} \right) + Q$$



$$0 = \rho c_p \dot{T}(r, t) - \frac{1}{r} \frac{\partial}{\partial r} \left(r k(T) \frac{\partial T(r, t)}{r} \right) - Q(r)$$

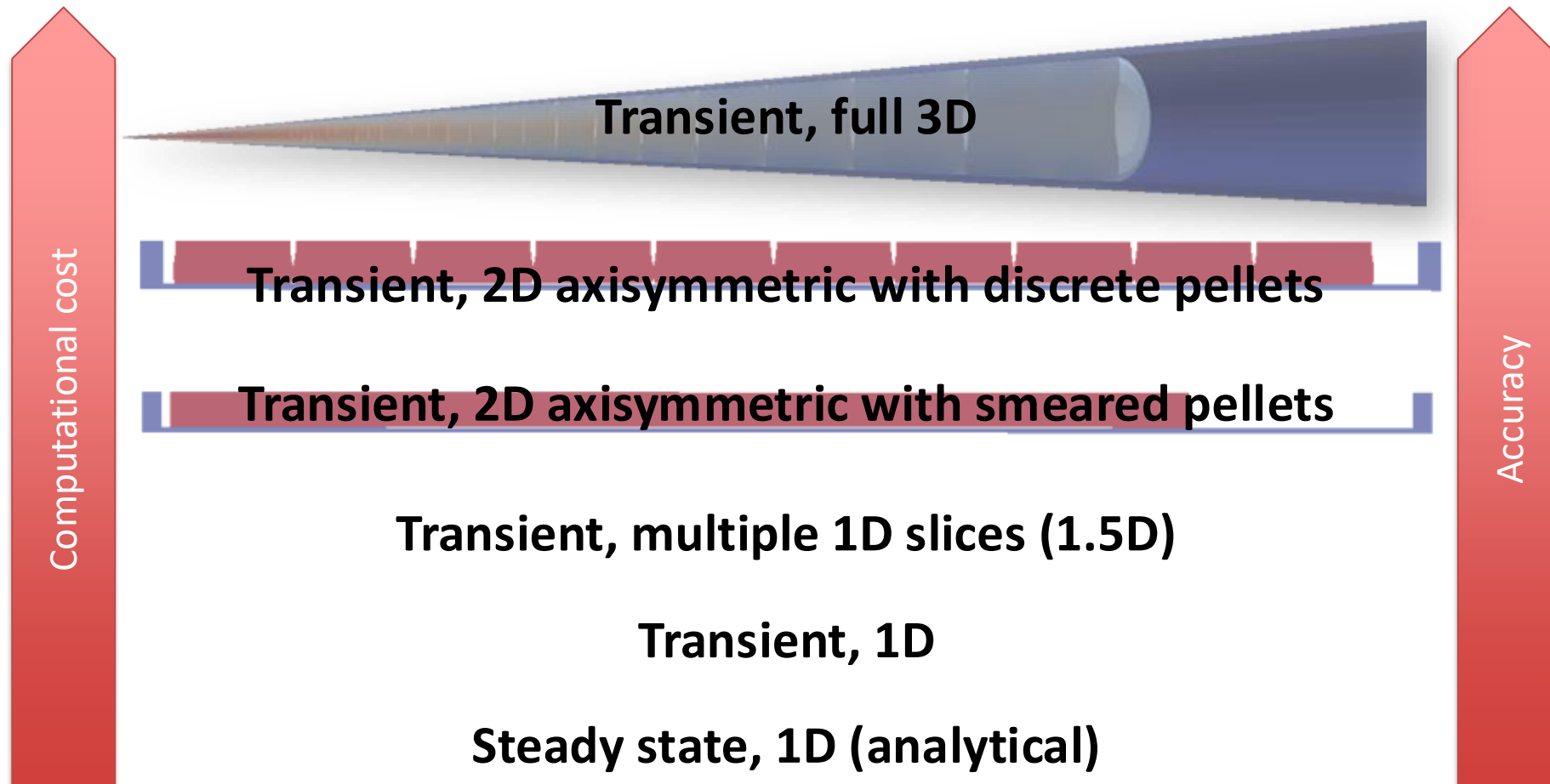
$$-\nabla \cdot \frac{\mathbf{K}}{\mu} \nabla p = 0 \in \Omega$$

$$(\nabla \psi, \frac{\mathbf{K}}{\mu} \nabla p) - \langle \psi, \frac{\mathbf{K}}{\mu} \nabla p \cdot \hat{n} \rangle = 0$$

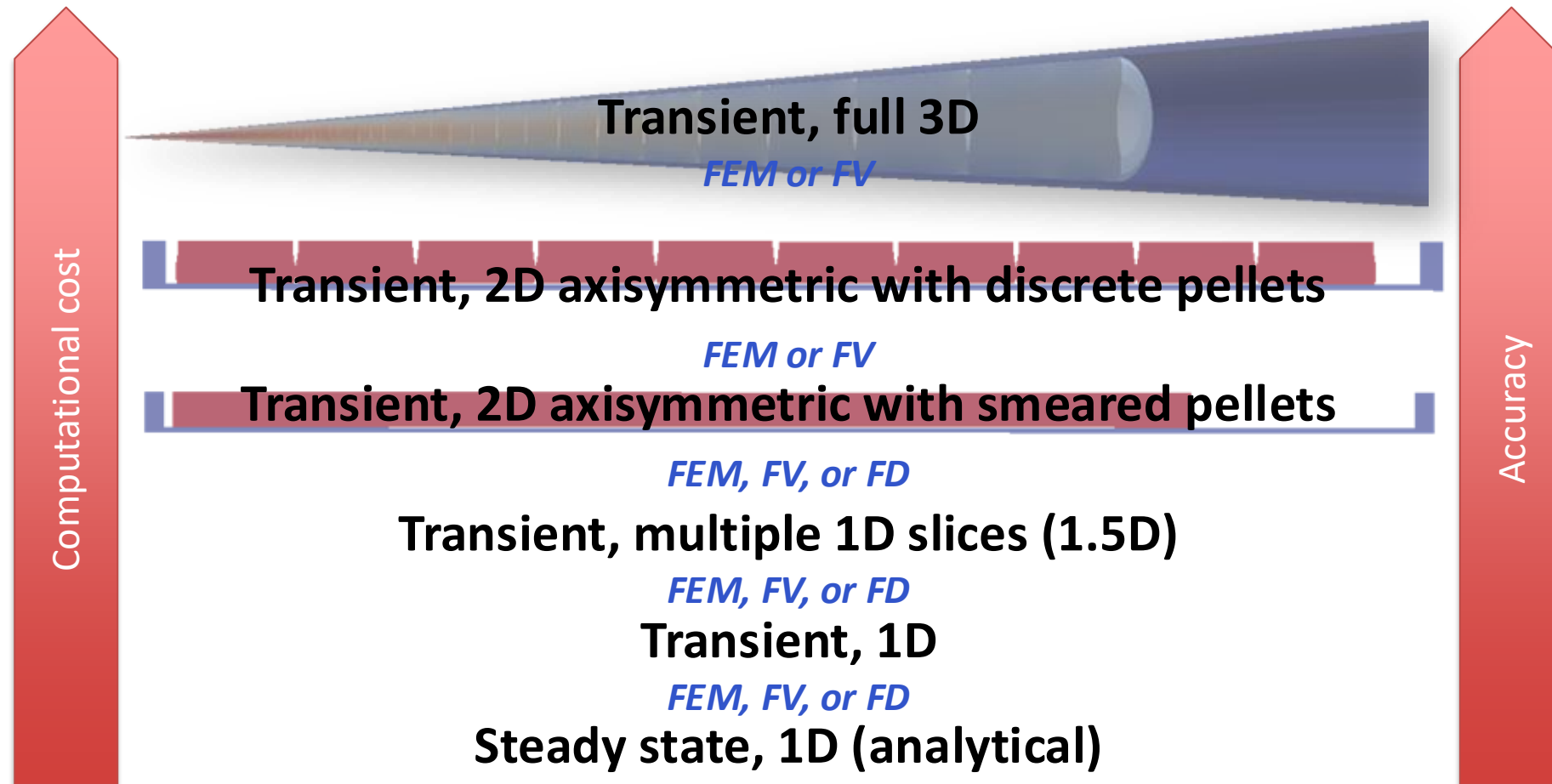
Spatial resolution

- Finite difference
 - Advantages
 - Simple
 - Easy to code
 - Fast
 - Disadvantages
 - Difficult to model complex geometries
 - Difficult to model complex BCs
 - Only represents solution at points
 - Difficult to represent heterogeneous properties
- Finite Volume
 - Advantages
 - Can model any geometry
 - Naturally conservative
 - Heterogeneous properties
 - Disadvantages
 - Boundary conditions add complexity
 - More complicated than finite difference
- Finite Element
 - Advantages
 - Can model any geometry
 - Can model any BC
 - Continuous representation
 - Heterogeneous properties
 - Disadvantages
 - Complicated
 - Somewhat more expensive

Different Fuel Performance Problems



Numerical Approaches to Different Fuel Performance Problems



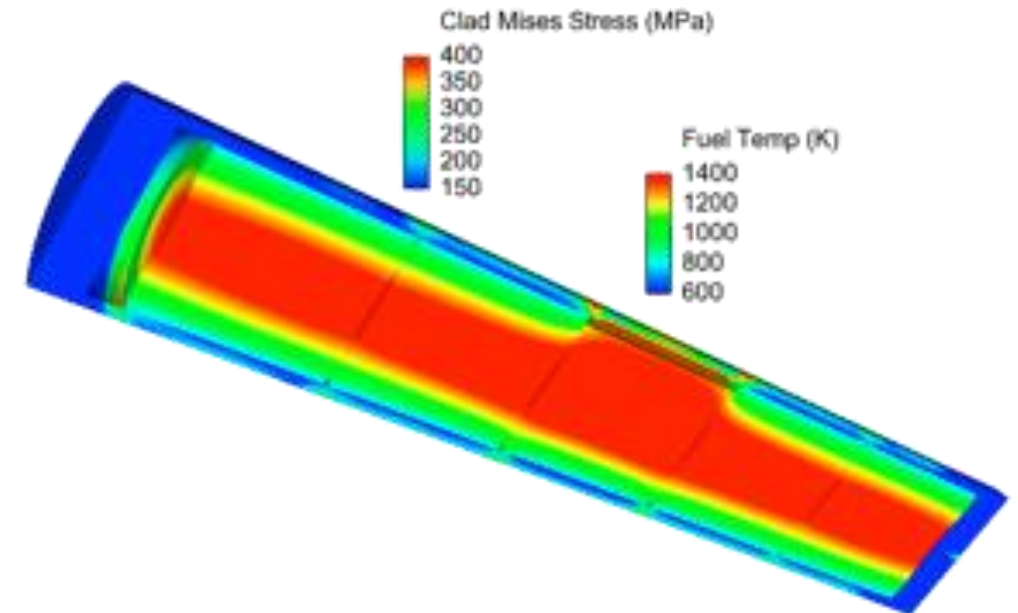
Heat equation solution approach summary

Approach	Solution	Assumptions
1D steady state	Analytical	Steady state, axisymmetric, no axial variation, constant k
1D transient	FEM, FD, FV	Axisymmetric, no axial variation
1.5D transient	FEM, FD, FV with multiple slices	Axisymmetric, no axial variation
2D transient, smeared pellets	FEM, FD, FV	Axisymmetric, fuel pellets act as one body, fuel pellets are perfect cylinders
2D transient, discrete pellets	FEM, FV	Axisymmetric
3D transient	FEM, FV	You have a big computer

Each numerical solution can be solved explicitly or implicitly

Solving with fuel performance codes

- Fuel performance codes primarily use either finite difference or finite element
- The earliest fuel performance codes solved the heat equation in 1.5D using finite difference (with multiple axial slices)
- More modern codes have switched to finite element, due to more flexibility with geometry and boundary conditions
- Finite volume is becoming an option



Summary

- The heat equation can be solved using numerical methods
- Explicit or implicit time integration can be used
- Spatial derivative solution methods divide the domain up into smaller pieces
 - Finite difference
 - Finite volume
 - Finite element
- Each discretization has strengths/weaknesses
- Finite element is primary method for high fidelity fuel performance codes

End of Module 1

- Problem session on Thursday 29th
- Exam on Feb 3rd
- Exam to be completed during class period
 - bring calculators, can use your own paper or what I provide
- Reminder of office hours on Wednesdays at 9am
- Reminder of slack channel for rapid communication
- Will schedule exams with distance students

- Will provide equation sheet for students in class. Will share equation sheet before hand.